Status Note: The Converse for Locally Finite Metric Universality

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Status. This is a `literature_already_answered` packet. The source question from Baudier–Lancien [1] is explicitly answered in Baudier's later paper [2], using an argument credited there to G. Schechtman.

Source question. Baudier–Lancien prove in Section 2, the theorem labeled `main` in [1], that there is a universal constant $C > 1$ such that whenever a Banach space X uniformly contains the spaces ℓ_∞^n, every locally finite metric space M admits a metric embedding $M \overset{C}{\hookrightarrow} X$. After their Proposition/Lemma showing that pointwise embeddability of all locally finite metric spaces is equivalent to uniformly bounded distortion, they ask in the final remark whether the converse of that theorem is true: if every locally finite metric space metrically embeds into a Banach space X, must X uniformly contain the ℓ_∞^n's?

Later answer. Baudier [2], Section 4, first proves a proper-metric-space converse: for $p = \infty$, Schechtman's argument gives that X has no nontrivial cotype if and only if all proper metric spaces admit suitable coarse bi-Lipschitz embeddings into X with universal constants. The proof uses finite grids

$$N_k = k^{-1}\mathbb{Z}^n \cap kB_{\ell_\infty^n},$$

passes to an ultrapower of X, applies the Heinrich–Mankiewicz differentiability principle to obtain linear copies of ℓ_∞^n, and then uses finite representability to return to X.

Most importantly for the source question, the final remark of [2] states the exact locally finite consequence: the converse of Baudier–Lancien's Theorem 2.1 holds, namely a Banach space X uniformly contains the ℓ_∞^n's if and only if every locally finite metric space Lipschitz embeds into X.

Scope and classification. This completely answers the converse question in [1] for the same Lipschitz/metric embedding notion used there. Since the supporting paper explicitly identifies the Baudier–Lancien theorem and states the converse, this packet is classified as `literature_already_answered`, not as a new full solution from the run.

References

[1] F. Baudier and G. Lancien, *Embeddings of locally finite metric spaces into Banach spaces*, Proc. Amer. Math. Soc. 136 (2008), 1029–1033; arXiv:math/0702266.

[2] F. Baudier, *Embeddings of proper metric spaces into Banach spaces*, Houston J. Math. 38 (2012), no. 1, 209–223; arXiv:0906.3696.